CIT 140 · Week 7 · Data-Quality Report

Meridian Forge · fictional inspection export v1

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Import status	20 rows loaded

Amber cells: your responses. Other results calculate from Prepared and Audit.

Rule	Check	Rows flagged	Share of all rows	**Affected Row_IDs**
Q01	Machine ID is required	2	10%	R03, R14
Q02	Work order is unique	2	10%	R05, R06
Q03	Status uses the agreed list	2	10%	R08, R16
Q04	Duration is in range	3	15%	R10, R18, R20
Q05	READY identifies an inspector	3	15%	R03, R07, R12
Distinct rows with any issue		11	55%	Count each row once, even when several Q-rules flag it. Do not add the five counts.

Scope: all 20 supplied records. Q02 flags every row in a repeated Work_Order_ID group; extra duplicate copies are a different measure. Unresolved fields remain flagged, not guessed.

Source: authored CIT140 case W07-S01 (Raw) and W07-S02 (rules). Report a data-quality finding, not a machine safety clearance. Explain your evidence and next action on page 2.

Week 7 · Explanation and refresh evidence

Finding: Row_ID, Q-number, actual value, source and effect

R03 is READY but is missing Machine_ID and Inspector_ID (Q1, Q5) and this means the record doesn't recognize machine, and there is no accountable inspector. R14 has no MACHINE_ID as well. R05 and R06 have the same work order which is unresolved in why there are duplicates of work orders(Q2). R10, R18, and R20 all have issues in Q4, where R10 has a time duration of 0, R20 is missing a time duration, and R18 has a time duration of 75 minutes. R7 and R12 have issues in Q5 in which they have incomplete inspector ID's. R08 and R16 have issues in Q3 where they have inccorect statuses where R8 has an incorrect spelling of READY and R16 has a status of done, neither of these are correct for their category.

Next action: what to verify, who owns it, and how to retest

For machine ID or inspection time checks maintenance records owner would need to check R03, R14, R10, r18, and R20 against the orginal. Since the shift supervisor checks the READY status or inspector assignment needs checking, they would need to check R03, R07, R12, R08, and R16. After corrections have been made, then they can be retested.

Limit: why you cannot guess a correction or prove accuracy

You cannot guess a correction for missing machine_id or inspector_id because it is dependent on the required specialist needed for the work order t o be completed which that information is not available in this export.

Professional connection: one framework idea and how you used it

One framework idea is using metrics to perform an initial data quality assessment which is essentially understanding the rules in which the values correspond to in order to be able to see an errors within the data which was beneficial in understanding in why the affected rows were flagged.

Refresh test: before, changed value, after, and restored result

Q4 before: rows flagged was 3 and share of all rows was 15%
Q4 after change of value and data refresh: rows flagged was 4 and was 20% share of all rows
Q4 after changing back from 0 to 20: changed back to as before, rows flagged in Q4 was 3, and 15% of all rows

Export only Report: two landscape pages, page break before row 26. Review both pages.